Zoo501

Quiz no 1

Due Date 5 June to 7 June 2023

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1. Sperm consist of… .. *All of these*
2. *Moy and Vacquier* demonstrated that bindin is located specifically on the acrosomal process exactly where it should be for sperm-egg recognition.
3. Miller found that when sperm were added to oocytes that had not yet completed their *2* meiotic division, there was no attraction of sperm to eggs.
4. Bleil and Wassarman *(1988)* isolated an 83-kDa glycoprotein, ZP3, from the mouse zona that was the active site for sperm binding.
5. The volume of a sea urchin egg is about 200 *picoliters*
6. In egg, Outside the plasma membrane is the vitelline envelope, which forms a fibrous mat around the egg. This envelope contains at least *eight* different glycoproteins and is often involved in sperm egg recognition.
7. The innermost layer of cumulus cells, immediately adjacent to the zona pellucida, is called the *corona radiata.*
8. In capacitation,protein phosphorylation occurs?*Fourth*
9. It appears that the acrosome-reacted sperm transfer their binding from ZP3 to the adjacent *ZP2* molecules. After a mouse sperm has entered the egg, the egg cortical granules release their contents.
10. Wilcox and colleagues *(1995)* found that nearly all human pregnancies result from sexual intercourse during a 6-day period ending on the day of ovulation.
11. The molecular changes that account for capacitation are, of *4* types.
12. The zona pellucida plays *two* major roles during fertilization.
13. In capacitation, *Third* , the membrane potential of the sperm becomes more negative as potassium ions leave the sperm.
14. The major motor portion of the flagellum is called the *axoneme*.
15. The interaction of sperm and egg generally proceeds according to *five* basic step.
16. The *Resact* diffuses readily in seawater and has a profound effect at very low concentrations when added to a suspension of Arbacia sperm.

17. The Cortical granules are membrane-bound structures containing *proteolytic* enzymes.
18. The difference between the acrosomal reaction in sea urchins and mammals may be due to the thickness of the extracellular envelopes surrounding the egg. *True*
19. Many eggs contain ultraviolet filters and *DNA* repair enzymes that protect them from sunlight; some eggs contain molecules that potential predators find distasteful; and the yolk of bird eggs even contains antibodies.
20. The exocytosis of the acrosomal vesicle is caused by the *calcium* - mediated fusion of the acrosomal membrane with the adjacent sperm plasma membrane?
21. In capacitation, the *Second* particular proteins or carbohydrates on the sperm surface are lost during capacitation.
22. In many species, the meeting of sperm and egg is not a simple matter and faces…*2*… problems?
23. In capacitation, the *first* the fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract?
24. The egg white is known as the Yolk? *False*
25. Contact with egg jelly causes the..*Exocytosis*.. of the sperms acrosomal vesicle and the release of proteolytic enzymes that can digest a path through the jelly coat to the egg surface?
26. Using immunological techniques..*May and vacquier* demonstrated that bindin is located specifically on the acrosomal process exactly where should be for sperm-egg recognitions.
27. The… *zona pellucida* … is a glycoprotein matrix, which is synthesized and secreted by the growing oocyte?
28. The instructions for proteins made during early development are already packaged in the oocyte. It is estimated that the eggs of sea urchins contain 25,000 to 50,000 different types of…*mRNA* … .?
29. The second part of the acrosomal reaction involves the extension of the acrosomal process. This protrusion arises through the polymerization of globular actin molecules into actin filaments. *True*
30. By undergoing the acrosomal reaction directly on the zona, the sperm is able to concentrate its proteolytic enzymes to lyse a hole in this envelope. Indeed, sperm that undergo the acrosomal reaction before they reach the zona pellucida are unable to penetrate it. *True*

31. In most marine invertebrates, the acrosomal reaction has *two* components.
32. The mechanisms of chemotaxis differ among species. One chemotactic molecule, a 14-amino acid peptide called *resact*, has been isolated from the egg jelly of the sea urchin Arbacia punctulata.
33. The *ZP3* is the specific glycoprotein in the mouse zona pellucida to which the sperm bind.
34. The *egg jelly* factors that initiate the acrosomal reaction in sea urchins are often highly specific to each species.
35. The acrosomal protein mediating this recognition is called bindin. In *1977*, Vacquier and co-workers isolated this nonsoluble 30,500-Da (Dalton) protein from the acrosome of Strongylocentrotus purpuratus and found it to be capable of binding to dejellied eggs of the same species.
36. The mammalian egg is also surrounded by a layer of cells called the *cumulus*.
37. The region of globular actin molecules lies between the nucleus and the acrosomal vesicle in Sea urchins.
38. The exocytosis of the acrosomal vesicle is caused by the calcium -mediated fusion of the acrosomal membrane with the adjacent sperm plasma membrane? *True*
39. Protein synthesis is accomplished by ribosomes and *tRNA*
40. The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation. The requirement for capacitation varies from species to species. *True*